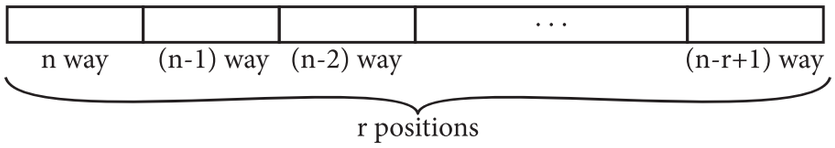
4.4.1 Permutations of distinct objects

In terms of function on any finite set say $S = \{x_1, x_2, ...x_n\}$, a permutation can be defined as a bijective mapping on the set S onto itself. The number of permutation on the set S is the same as the total number of bijective mappings on the set S.

We denote the number of permutations by ${}^nP_r$.

Theorem 4.1: If n, r are positive integers and $r \leq n$, then the number of permutations of n distinct objects taken r at a time is $n\,(n-1)\,(n-2)\cdots(n-r+1)$.

Proof. A permutation is an ordering. A permutation of n distinct objects taken r at a time is formed by filling of r positions, in a row with objects chosen from the given n distinct objects.



There are n objects that can be filled in the first position. For the second position there are remaining $n-1$ objects. There are $n-2$ objects for the third position. Continuing like this until finally we place one of the $(n-(r-1))$ possible objects in the rth position. By the rule of product we conclude ${}^nP_r = n\,(n-1)\,(n-2)\cdots(n-r+1)$.

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Theorem 4.2: If $n \geq 1$, and $0 \leq r \leq n$, then ${}^nP_r = \dfrac{n!}{(n-r)!}$.

Proof. By Theorem 4.1, we have,

$$\begin{aligned} {}^nP_r &= n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) \\ &= \frac{n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) \times (n-r) \times (n-r-1) \cdots 2 \times 1}{(n-r) \times (n-r-1) \times \cdots \times 2 \times 1} \\ &= \frac{n!}{(n-r)!}. \end{aligned}$$

■

To be specific, if n is a positive integer, and r is any non-negative integer, we can represent,

$${}^nP_r = \begin{cases} \dfrac{n!}{(n-r)!} & \text{for } r \leq n, \\ 0 & \text{for } r > n. \end{cases}$$

$${}^nP_r = \begin{cases} {}^nP_n = n! & \text{for } r = n, \\ {}^nP_0 = 1 & \text{for } r = 0. \end{cases}$$

The n different objects arranged in a row is ${}^nP_n = n!$ ways.

Theorem 4.3: The number of permutations of n different objects taken r at a time where repetition is allowed, is n^r.